

Streptococcal Pharyngitis (MONDO:0021783): A Comprehensive Disease Characteristics Report

Category: Infectious Disease · Evidence base: 55 papers reviewed, 12 confirmed findings

Summary

Streptococcal pharyngitis is an acute, usually self-limited infectious tonsillopharyngitis caused by Group A Streptococcus (*Streptococcus pyogenes*, GAS). The organism — a Gram-positive, catalase-negative, β -hemolytic coccus growing in chains — colonizes the pharyngeal epithelium of the upper respiratory tract, which is a major human reservoir. Adhesion and colonization, mediated by virulence factors including M protein, streptokinase, and pili (T-antigen), are the initiating mechanistic steps. The infection produces a characteristic clinical picture of abrupt fever, severe sore throat, odynophagia, tonsillar exudate, palatal petechiae, a swollen red uvula, and tender anterior cervical lymphadenopathy, typically in children aged 5–15 years.

The disproportionate clinical importance of this common illness derives not from the acute pharyngitis itself — which resolves in most patients within a week — but from its **complications**. These fall into two branches: (1) *suppurative* complications from local/contiguous spread (peritonsillar and retropharyngeal abscess, otitis media, sinusitis, mastoiditis, and rarely intracranial extension); and (2) *non-suppurative autoimmune sequelae* driven by molecular mimicry between streptococcal antigens (particularly M protein) and human tissue — acute rheumatic fever (ARF), rheumatic heart disease (RHD), acute post-streptococcal glomerulonephritis (APSGN), and neuropsychiatric sequelae (Sydenham chorea; the proposed PANDAS entity). Only a small fraction of infections progress to these sequelae, and this progression is gated by host genetics, chiefly HLA class II alleles.

Clinically, **diagnosis requires microbiological confirmation** (rapid antigen detection test [RADT], throat culture, or molecular point-of-care test) rather than clinical scoring alone, because GAS pharyngitis is phenotypically indistinguishable from viral and non-group-A

streptococcal causes. **Treatment with penicillin or amoxicillin — to which GAS remains universally susceptible** — provides modest symptomatic benefit but importantly reduces suppurative complications and lowers ARF incidence by 70–80%. The autoimmune sequelae are diagnosed by dedicated frameworks (the 2015 revised Jones criteria for ARF, with Doppler echocardiography), and secondary benzathine penicillin G prophylaxis prevents progression of latent RHD (GOAL randomized controlled trial). RHD affects more than 40.5 million people worldwide and causes ~306,000 deaths annually, making primary and secondary prevention of streptococcal pharyngitis a global public-health priority.

Disease Information (Section 1)

Streptococcal pharyngitis (also "strep throat," GAS/GABHS pharyngitis, streptococcal tonsillopharyngitis, streptococcal sore throat) is an acute bacterial infection of the pharynx and tonsils caused by *Streptococcus pyogenes*. Key identifiers:

Resource	Identifier
Mondo	MONDO:0021783
ICD-10	J02.0 (streptococcal pharyngitis); J03.00 (acute streptococcal tonsillitis)
ICD-11	CA02.0
MeSH	Pharyngitis (D010612); <i>Streptococcus pyogenes</i> (D013297); Streptococcal Infections (D013290)
NCBI Taxonomy	<i>Streptococcus pyogenes</i> , txid1314
SNOMED CT	43878008 (streptococcal sore throat)

The information for this disease is derived from **aggregated disease-level resources** — clinical guidelines, Cochrane systematic reviews, epidemiological studies, and mechanistic literature — rather than individual patient EHR records (though some cited cohorts, e.g.,

P 34805428

are EHR-based). OMIM and Orphanet do not assign a primary entry to streptococcal pharyngitis itself (it is not a Mendelian disorder), though related host-susceptibility phenotypes for rheumatic fever/RHD appear in the genetics literature.

Key Findings

F001 — *Streptococcus pyogenes* (Group A Streptococcus) is the causal agent

GAS is a Gram-positive, catalase-negative, β -hemolytic coccus that grows in chains and colonizes the upper respiratory tract, which serves as a major reservoir. Bacterial **adhesion and colonization of the pharyngeal epithelium are the initiating steps** of infection, mediated by virulence factors including M protein and streptokinase.

"The upper respiratory tract and skin are major reservoirs for GAS infections. The ability of GAS to establish an infection in the new host at these anatomical sites primarily results from two distinct physiological processes, namely bacterial adhesion and colonization." — PMID: 27312939

"It is known to cause a wide range of infections in children, ranging from mild upper respiratory tract infections, such as pharyngitis, to severe invasive disease." — PMID: 40572286

Ontology suggestions: infectious agent NCBITaxon:1314 (*S. pyogenes*); GO:0007155 (cell adhesion); UBERON:0006562 (pharynx); CL:0000066 (epithelial cell).

F002 — Post-streptococcal autoimmune sequelae (ARF, RHD, APSGN) arise via molecular mimicry

Acute rheumatic fever is an autoimmune disorder that follows GAS pharyngitis or impetigo in children and adolescents and may evolve into rheumatic heart disease with persistent valve damage. GAS triggers post-infectious sequelae — APSGN, ARF, and RHD — attributed largely to **molecular mimicry** between streptococcal antigens (e.g., M protein) and human tissue proteins. Critically, detection and treatment of streptococcal pharyngitis reduces ARF incidence by 70–80%, confirming the causal link.

"Acute rheumatic fever (ARF) is an autoimmune disorder resulting from Group A Streptococcus (GAS) pharyngitis or impetigo in children and adolescents, which may evolve to rheumatic heart disease (RHD) with persistent cardiac valve damage." — PMID: 40484016

"GAS also notably triggers post-infectious immune sequelae, including acute poststreptococcal glomerulonephritis (APSGN), acute rheumatic fever (ARF), and rheumatic heart disease (RHD), which are major health burdens, especially in low-income countries." — PMID: 40572286

"Detection and treatment of streptococcal pharyngitis in children reduces acute rheumatic fever by 70-80%." — PMID: 41956705

Ontology suggestions: GO:0002250 (adaptive immune response); GO:0042110 (T cell activation); MONDO:0005295 (rheumatic heart disease); UBERON:0002007 (heart valve).

F003 — Diagnosis requires microbiological confirmation; clinical scores are insufficient alone

Children with GABHS pharyngitis typically present with abrupt fever, intense throat pain, odynophagia, an inflamed pharynx, enlarged erythematous tonsils, a red swollen uvula, and tender anterior cervical lymphadenopathy. Because these features overlap with viral causes, suspected cases **should be confirmed by microbiological testing** (culture, RADT, or molecular point-of-care test) before starting antibiotics. Clinical scores perform poorly: in a pediatric cohort, a Centor score ≥ 3 had only 22.3% sensitivity and 79.0% specificity for RADT positivity, and 17.1% of RADT-negative patients had positive throat cultures.

"Children with GABHS pharyngitis typically present with an abrupt onset of fever, intense pain in the throat, pain on swallowing, an inflamed pharynx, enlarged and erythematous tonsils, a red and swollen uvula, enlarged tender anterior cervical lymph nodes." — PMID: 37493159

"Patients suspected of having GABHS pharyngitis should be confirmed by microbiologic testing (e.g., culture, rapid antigen detection test, molecular point-of-care test) of a throat swab specimen prior to the initiation of antimicrobial therapy." — PMID: 37493159

"The Centor score alone does not seem to be of any utility in guiding the diagnosis of suspected streptococcal pharyngitis." — PMID: 39528865

Supporting this, a systematic review and meta-analysis of McIsaac and Centor scores concluded both are "equally ineffective at triaging patients who need antibiotics" ([PMID: 38182052](#)). Cough and coryza are useful to *rule out* GAS; adding RADT to a Centor 3–4 raises positive predictive value to ~93% ([PMID: 34535115](#)). In college students, GAS and non-group-A streptococcal pharyngitis were clinically indistinguishable ([PMID: 34805428](#)).

Ontology / lab-test suggestions: HP:0025439 (pharyngitis); LOINC 626-2 (throat culture); LOINC 60489-2 (*S. pyogenes* rapid Ag).

F004 — GAS remains universally penicillin-susceptible; antibiotics give modest symptomatic benefit but prevent complications

GAS remains universally susceptible to penicillin, though macrolide and lincosamide resistance is increasing among invasive isolates, with uncertain clinical consequences. Antibiotics provide only modest benefit for sore-throat symptoms, but effectiveness increases in patients with GABHS-positive swabs.

"GAS remains universally susceptible to penicillin but there are increasing reports of macrolide and lincosamide resistance, particularly in invasive isolates, with uncertain clinical consequences." — [PMID: 39259691](#)

"Antibiotics provide only modest benefit in treating sore throat, although their effectiveness increases in people with positive throat swabs for group A beta-haemolytic streptococci (GABHS)." — [PMID: 37965935](#) (Cochrane)

The Cochrane comparative-antibiotics review (19 trials, 5,839 participants) found **no clinically relevant differences** between cephalosporins/macrolides and penicillin for symptom resolution; given low cost and absence of resistance, penicillin remains first-line ([PMID: 33728634](#)). Where penicillin V is unavailable, amoxicillin 50 mg/kg/day for 10 days is first choice, with macrolides reserved for type-I penicillin allergy ([PMID: 22691611](#)).

Ontology suggestions (NCIT/CHEBI): NCIT:C61785 (Penicillin); NCIT:C287 (Amoxicillin); CHEBI:17334 (penicillin); CHEBI:2676 (amoxicillin).

F005 — Host HLA class II alleles modulate susceptibility to post-streptococcal rheumatic fever/RHD

Only a small fraction of GAS pharyngitis episodes progress to rheumatic carditis, implicating host genetics. A meta-analysis (13 studies; 1,065 patients / 1,691 controls) identified **HLA-DRB1*07 as a susceptibility allele (OR $\approx$ 1.68)** and **HLA-DRB1*15 as protective**. GWAS implicate the HLA-DQA1–HLA-DQB1 region and the immunoglobulin heavy-chain locus (IGHV4-61) on chromosome 14. Twin studies show much higher RF concordance in monozygotic (19%) than dizygotic (2.5%) twins. TNFA-308 and mannose-binding lectin (MBL) variants are also associated.

*"The results of the meta-analysis suggest that the differential presentation of autoimmune peptides by HLA-DRB1*07 (susceptible) and HLA-DRB1*15 (protective) alleles with different affinities may play a crucial role in the pathogenesis of RF/RHD."* — [PMID: 32967480](#)

"Early findings implicate not only HLA, particularly the HLA-DQA1 to HLA-DQB1 region, but also the immunoglobulin heavy chain locus, including the IGHV4-61 gene segment, on chromosome 14." — [PMID: 31519994](#)

"the high concordance rate for RF in monozygotic twins (19%) compared to dizygotic twins (2.5%), and the high familial incidence of RF suggest the involvement of host genetic factors in susceptibility to RF" — [PMID: 37406855](#)

An independent HLA study found HLA-DRB1*13, DRB5*, and DRB3* to be protective against rheumatic valve damage ([PMID: 16426242](#)); MBL deficiency and TNFA-308 associations were reviewed in [PMID: 17804571](#). **HGNC genes:** HLA-DRB1, HLA-DQA1, HLA-DQB1, IGHV4-61, TNF, MBL2.

F006 — Streptococcal superantigens (SpeA) link pharyngitis to scarlet fever and subvert immunity; the M1UK clone is emerging

Streptococcal pyrogenic exotoxin A (SpeA) expression is epidemiologically linked to tonsillopharyngitis and scarlet-fever outbreaks. As a superantigen, SpeA drives massive tonsil T-cell proliferation with proinflammatory cytokine release, yet paradoxically causes **B-cell apoptosis and abrogation of IgA/IgM/IgG production**, subverting protective humoral immunity. The emergent **M1UK variant** (27 SNPs distinct from M1global) is characterized by increased *speA* expression and is driving surges in scarlet fever and invasive disease.

"Streptococcal pyrogenic exotoxin (Spe) A expression is epidemiologically linked to streptococcal tonsillo-pharyngitis and outbreaks of scarlet fever" — PMID: 30815853

"marked B cell apoptosis and abrogation of total immunoglobulin (Ig)A, IgM, and IgG production occurred in the presence of SpeA and other superantigens" — PMID: 30815853

"M1UK differs from progenitor M1global genotype by 27 single-nucleotide polymorphisms and is characterized by increased speA superantigen expression in vitro." — PMID: 39206960

Genomic surveillance in Shanghai documented a parallel rise in *emm1* isolates carrying *speA*, *speC*, and *spd1* virulence genes and *ermB/tetM* resistance determinants (PMID: 32562850). GO: GO:0050852 (T cell receptor signaling); GO:0006915 (apoptotic process).

F007 — Lewis rat autoimmune valvulitis (RAV) model recapitulates M-protein-driven rheumatic carditis

Immunization of Lewis rats with recombinant GAS M5 protein induces mitral valvulitis and myocarditis with CD3⁺/CD4⁺/CD68⁺ mononuclear infiltration of valve tissue and P-R-interval prolongation on ECG — features resembling human rheumatic carditis. Both anti-M5 antibodies and M5-specific T cells transfer carditis to naïve syngeneic rats, and repeat M-protein exposure exacerbates cardiac damage via an enhanced **anti-cardiac-myosin antibody** response. An M5 B-repeat peptide (aa 161–180) induces lymphocytes cross-reactive to cardiac myosin.

"serum plus in vitro expanded rM5-specific T-cells from hyperimmune rats were capable of transferring carditis to naïve syngeneic animals" — PMID: 31062619

"repetitive booster immunization with GAS-derived recombinant M protein (rM5) resulted in an enhanced anti-cardiac myosin antibody response that may contribute to the breaking of immune tolerance leading to RF/RHD" — PMID: 27562362

"Rats immunized with streptococcal M5 protein developed valvular lesions, distinguished by infiltration of CD3(+), CD4(+), and CD68(+) cells into valve tissue" — PMID: 19273562

This model directly demonstrates the molecular-mimicry mechanism (F002) is *pathogenic and transferable*. Detailed induction protocols are consolidated in PMID: 42261692. **CL terms:** CL:0000084 (T cell), CL:0000624 (CD4⁺ T cell), CL:0000235 (macrophage, CD68⁺).

F008 — GAS pharyngitis causes suppurative and non-suppurative complications; antibiotics reduce them

Suppurative complications (local spread) include peritonsillar/retropharyngeal abscess (quinsy), sinusitis, mastoiditis, otitis media, and rarely orbital cellulitis, meningitis, brain abscess, and intracranial venous sinus thrombosis. **Non-suppurative (autoimmune)** complications include ARF, APSGN, and neuropsychiatric sequelae (Sydenham chorea; the proposed PANDAS entity with anti-D1R dopamine-receptor autoantibodies targeting the basal ganglia). A Cochrane meta-analysis (27 trials, 2,835 cases) found antibiotics reduced acute otitis media (RR 0.30, 95% CI 0.15–0.58), quinsy, and ARF by more than two-thirds (RR 0.22, 95% CI 0.02–2.08).

"Complications associated to group-A streptococcal pharyngitis include non-suppurative complications such as acute rheumatic fever and glomerulonephritis and suppurative complications such as peritonsillar or retropharyngeal abscess, sinusitis, mastoiditis, otitis media, meningitis, brain abscess, or thrombosis of the intracranial venous sinuses." — PMID: 23067784

"Antibiotics reduced the incidence of acute otitis media (RR 0.30; 95% CI 0.15 to 0.58)" — PMID: 17054126

"Emerging molecular evidence identifies anti-D1R autoantibodies, acting via G protein-and beta-arrestin-mediated signalling, as candidate bi[omarkers]" — PMID: 42196589

The PANDAS/PANS construct remains controversial but biologically plausible; single-cell RNA-seq of a Sydenham chorea patient showed B-cell HLA-DR/DQ upregulation and plasma-cell proteasomal activation, supporting a B-cell-mediated autoantibody hypothesis (PMID: 40859454; PMID: 42603022). **HP terms:** HP:0000388 (otitis media), HP:0000246 (sinusitis), HP:0002072 (chorea for Sydenham).

F009 — GAS pili (T-antigen) mediate adhesion/colonization and are leading vaccine candidates

GAS pili are surface-exposed structures involved in adhesion and colonization; the major component is the **T-antigen**, which multimerizes to form the pilus shaft and is encoded in the FCT genomic region. A multivalent T-antigen fusion vaccine (TeeVax1–3; 18 T-antigens) produced opsonophagocytic, cross-reactive antibodies covering ~95% of 21 T-antigens and

conferred protection against invasive disease in mice. Mucosal delivery of GAS pili via *Lactococcus lactis* generated neutralizing/opsonophagocytic antibodies and improved nasopharyngeal GAS clearance.

"Pili of Group A Streptococcus (GAS) are surface-exposed structures involved in adhesion and colonisation of the host during infection. The major protein component of the GAS pilus is the T-antigen, which multimerises to form the pilus shaft." — PMID: 33623073

"Combining TeeVax1-3 produced a robust antibody response in rabbits that was cross-reactive to a full panel of 21 T-antigens, expected to provide over 95% vaccine coverage." — PMID: 33623073

"intranasal immunisation of mice improved clearance rates of GAS after nasopharyngeal challenge" — PMID: 28775292

No licensed GAS vaccine yet exists; pilus/T-antigen and M-protein-based approaches are the leading strategies (PMID: 38543606). GO: GO:0009289 (pilus); GO:0007155 (cell adhesion).

F010 — The ARF sequela is diagnosed by the 2015 revised Jones criteria (AHA)

The 2015 AHA revision of the Jones criteria is the international gold standard for diagnosing ARF. It **stratifies by population risk**, offering two diagnostic pathways — prioritizing specificity in low-risk and sensitivity in moderate/high-risk populations — recommends Doppler echocardiography in all suspected/confirmed cases, and allows **subclinical carditis** to fulfill a major criterion. Evidence of antecedent GAS infection (elevated/rising ASO or anti-DNase B, or positive throat culture/RADT) is a required supporting criterion.

"update those criteria to also take into account recent evidence supporting the use of Doppler echocardiography in the diagnosis of carditis as a major manifestation of acute rheumatic fever" — PMID: 25908771

"the criteria consider the risk within a population and offer two separate diagnostic pathways that prioritise specificity among those at low risk and sensitivity among those at moderate/high risk" — PMID: 27326214

Lab biomarkers: anti-streptolysin O (ASO), anti-DNase B. **Imaging:** Doppler echocardiography.

F011 — APSGN presents as immune-complex nephritic syndrome with hypocomplementemia

APSGN is an acute autoimmune kidney condition triggered by pharyngitis or skin infection with **specific nephritogenic *S. pyogenes* strains**; it is the most common cause of pediatric acute glomerulonephritis globally. Children present with a nephritic picture: edema, painless hematuria, and hypertension. In a 100-child cohort, low serum C3 occurred in 93.5%, elevated anti-DNase B in 100%, and elevated ASO in 85%; biopsy showed post-infectious nephritis with immune-complex deposits (some crescentic). Treatment is largely supportive (managing hypertension/fluid overload). Incidence is strongly tied to childhood socioeconomic disadvantage and dropped after COVID-19 non-pharmaceutical interventions.

"Acute post-streptococcal glomerulonephritis (APSGN) is an acute autoimmune kidney condition triggered by skin infection or pharyngitis caused by specific strains of Streptococcus pyogenes (Group A streptococcus)." — PMID: 40174621

"Children typically present with a nephritic clinical picture: oedema, painless haematuria and hypertension." — PMID: 40174621

"C3 levels were low in 86/92 (93.5%) children; 94/94 (100%) children had elevated anti-deoxyribonuclease-B (anti-DNase-B) levels; and 80/94 (85%) also had elevated anti-streptolysin O titre (ASOT) at presentation" — PMID: 38170231

APSGN hospitalizations fell markedly after COVID-19 NPIs (PMID: 38688264); in Nepal, C3 was depressed in 61.9–100% of cases and pyoderma was the dominant preceding route (PMID: 40119285). An experimental rabbit study proposed that GAS IgG-binding surface proteins trigger anti-IgG immune-complex formation and glomerular deposition, offering a complementary (non-mimicry) pathogenic route for APSGN (PMID: 15676011). **HP terms:** HP:0000790 (hematuria), HP:0000969 (edema), HP:0000822 (hypertension), HP:0000093 (proteinuria); **UBERON:0000074** (renal glomerulus).

F012 — Secondary benzathine penicillin G prophylaxis prevents progression of latent RHD (GOAL RCT)

The **GOAL randomized controlled trial** (Uganda; 916 children/adolescents aged 5–17 with echocardiographically confirmed latent RHD) compared intramuscular penicillin G benzathine every 4 weeks for 2 years versus no prophylaxis. Echocardiographic progression at 2 years was significantly lower in the prophylaxis arm, establishing that secondary antibiotic prophylaxis prevents progression of screen-detected latent RHD.

"Participants were randomly assigned to receive either injections of penicillin G benzathine (also known as benzathine benzylpenicillin) every 4 weeks for 2 years or no prophylaxis." — PMID: 34767321

"Rheumatic heart disease affects more than 40.5 million people worldwide and results in 306,000 deaths annually." — PMID: 34767321

Natural-history data show untreated moderate-to-severe latent RHD progresses in ~47.6% of cases, with younger age and morphological mitral-valve features as risk factors ([PMID: 28972003](#)). The GOAL protocol powered for a 50% relative risk reduction, randomizing 916 participants ([PMID: 31301533](#)). **NCIT:** benzathine benzylpenicillin (NCIT:C47476).

Mechanistic Model / Interpretation

Ordered causal chain

1. GAS is transmitted via respiratory droplets and CONTACTS pharyngeal epithelium.
2. Surface adhesins (M protein, pili/T-antigen, fibronectin-binding proteins)
MEDIATE adhesion → LEADS TO colonization of the tonsillar/pharyngeal mucosa. [F001, F009]
3. Colonization + secreted toxins (streptolysins, SpeA superantigen, proteases)
TRIGGER local innate inflammation → RESULTS IN the acute pharyngitis phenotype
(fever, sore throat, tonsillar exudate, tender cervical nodes). [F001, F003, F006]
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 - | -- BRANCH A (suppurative): unchecked local spread LEADS TO peritonsillar/
retropharyngeal abscess, otitis media, sinusitis, mastoiditis, and rarely
intracranial extension. [F008]
 - |
 - | -- BRANCH B (immune / non-suppurative):
 - 4B. Antigen presentation of streptococcal peptides (M protein, N-acetyl-
glucosamine) by susceptible HLA class II alleles (DRB1*07, DQA1/DQB1)
ACTIVATES cross-reactive CD4+ T cells and B cells. [F002, F005]
 - 5B. Molecular mimicry → cross-reactive antibodies + T cells RECOGNIZE
human cardiac myosin, valve endothelium, glomerular / neuronal antigens.
(Demonstrated & transferable in Lewis rat RAV model.) [F002, F007]
 - |
 - | -- Cardiac branch: valvulitis/carditis → RESULTS IN acute rheumatic
fever → repeated exposure LEADS TO chronic rheumatic heart disease. [F007]
 - | -- Renal branch: immune-complex deposition in glomeruli (nephritogenic
strains) → RESULTS IN APSGN nephritic syndrome + hypocomplementemia. [F011]
 - | -- Neuro branch (inferred/controversial): anti-neuronal / anti-D1R
autoantibodies target basal ganglia → Sydenham chorea / PANDAS. [F008]

Upstream vs downstream

Layer	Element	Direction
Initiating lesion	GAS adhesion/colonization of pharynx (M protein, pili)	Most upstream [F001, F009]
Amplifier	Superantigen (SpeA) T-cell activation; humoral subversion	Upstream–mid [F006]
Gatekeeper	Host HLA class II genotype (DRB1*07 susceptible; DRB1*15 protective)	Determines whether Branch B proceeds [F005]
Effector (autoimmune)	Cross-reactive antibodies + T cells vs cardiac myosin/valve/glomerulus/neurons	Downstream [F002, F007, F011]
Clinical endpoint	Pharyngitis (acute) → ARF/RHD, APSGN, chorea (weeks–years later)	Terminal

Cell types and processes

Pharyngeal/tonsillar **epithelial cells** (CL:0000066) are the primary infection site; tonsillar **CD4⁺ T cells** (CL:0000624) and **B cells** (CL:0000236) mediate the superantigen response and autoimmunity; **macrophages** (CL:0000235, CD68⁺) and **plasma cells** infiltrate target tissues. Key GO biological processes: GO:0007155 (cell adhesion), GO:0050852 (TCR signaling), GO:0002250 (adaptive immune response), GO:0006956 (complement activation, APSGN), GO:0006915 (apoptosis, superantigen-induced B-cell death). Anatomy: UBERON:0006562 (pharynx), UBERON:0002372 (tonsil), UBERON:0002007 (heart valve), UBERON:0000074 (glomerulus), UBERON:0002420 (basal ganglia).

Evidence Base

PMID	Role in report	What it supports
27312939	Primary	Pharynx as reservoir; adhesion/colonization as initiating steps (F001)
40572286	Primary	GAS causes pharyngitis; triggers APSGN/ARF/RHD sequelae (F001, F002)
40484016	Primary	ARF as autoimmune sequela evolving to RHD (F002)
41956705	Primary	Treating pharyngitis reduces ARF 70–80% (F002)
37493159	Review	Clinical phenotype; need for microbiological confirmation (F003)
39528865	Primary	Centor score inadequate alone (F003)
38182052	Meta-analysis	McIsaac/Centor equally ineffective for triage (F003)
34535115	Primary	Cough/coryza rule out GAS; RADT+Centor PPV 93% (F003)
34805428	EHR cohort	GAS vs non-group-A pharyngitis clinically indistinguishable (F003)
39259691	Review	Universal penicillin susceptibility; macrolide resistance (F004)
33728634	Cochrane	No antibiotic superior to penicillin; penicillin first-line (F004)
22691611	Guideline	Amoxicillin dosing; macrolides for penicillin allergy (F004)
32967480	Meta-analysis	HLA-DRB1*07 susceptible; DRB1*15 protective (F005)
31519994	GWAS review	HLA-DQ + IGHV4-61 loci (F005)
37406855	Primary	Twin concordance 19% vs 2.5% (F005)
16426242	Primary	DRB1*13/DRB5*/DRB3* protective (F005)
17804571	Review	TNFA-308, MBL deficiency, DR7-restricted M5 recognition (F005)
30815853	Primary	SpeA links to pharyngitis/scarlet fever; B-cell apoptosis (F006)
39206960	Primary	M1UK clone, 27 SNPs, increased speA (F006)
32562850	Primary	emm1 rise, speA/speC virulence genes (F006)
31062619	Model organism	M5 Ab/T cells transfer carditis (F007)
27562362		Repeat M-protein → anti-cardiac-myosin Ab (F007)

PMID	Role in report	What it supports
	Model organism	
19273562	Model organism	CD3/CD4/CD68 valve infiltration (F007)
23067784	Case/review	Suppurative + non-suppurative complication list (F008)
17054126	Cochrane	Antibiotics reduce otitis media RR 0.30 (F008)
42196589	Review	Anti-D1R autoantibodies in PANDAS (F008)
40859454	scRNA-seq	B-cell HLA-DR/DQ upregulation in Sydenham chorea (F008)
33623073	Primary	Pili/T-antigen; TeeVax ~95% coverage (F009)
28775292	Model organism	Mucosal pilus vaccine improves clearance (F009)
25908771	Guideline	2015 Jones criteria; echocardiography (F010)
27326214	Guideline	Risk-stratified diagnostic pathways (F010)
40174621	Review	APSGN definition + nephritic phenotype (F011)
38170231	Primary	Lab frequencies: C3 low 93.5%, anti-DNase B 100% (F011)
15676011	Model organism	IgG-binding proteins → immune complexes (APSGN/carditis) (F011)
34767321	RCT	GOAL: BPG prevents latent RHD progression; 40.5M affected (F012)
28972003	Cohort	Latent RHD natural history: 47.6% progression (F012)

Evidence-source mix: human clinical (guidelines, RCTs, cohorts, meta-analyses) dominates; model-organism data (Lewis rat RAV, mouse/rabbit vaccine and immune-complex studies) supports mechanism; in-vitro data (tonsil superantigen assays) and computational/genomic surveillance (M1UK, emm typing, GWAS) complete the picture.

Section-by-Section Coverage Notes

- **Etiology (2):** Infectious cause = GAS (F001). Genetic *risk modifiers* are host HLA class II (F005) — these gate autoimmune sequelae, not primary infection. Environmental risk factors: crowding, school age (5–15 y), winter/early-spring seasonality, socioeconomic disadvantage (esp. APSGN/RHD). Gene–environment interaction: susceptible HLA + GAS exposure is required for ARF (F002, F005).
- **Phenotypes (3):** Fever (HP:0001945), pharyngitis (HP:0025439), odynophagia/dysphagia (HP:0002015), tonsillar exudate, cervical lymphadenopathy (HP:0002751), palatal petechiae, scarlatiniform rash (scarlet fever). Childhood onset; mild–moderate, self-limited/episodic. Sequela phenotypes: carditis (HP:0001635), chorea (HP:0002072), hematuria (HP:0000790).
- **Genetic/molecular (4):** No causal *human* gene (not Mendelian). Susceptibility loci: HLA-DRB1, HLA-DQA1/DQB1, IGHV4-61, TNF, MBL2 (F005). Pathogen genetics: emm/M-protein types, speA, M1UK 27-SNP variant (F006).
- **Environmental (5):** Respiratory droplet transmission; crowding; infectious agent NCBITaxon:1314. No toxin/occupational etiology.
- **Anatomy (7):** Primary — pharynx/tonsils (UBERON:0006562, UBERON:0002372); epithelial tissue. Secondary — heart valves (mitral > aortic), renal glomeruli, basal ganglia, middle ear/sinuses.
- **Temporal (8):** Acute onset; self-limited over ~1 week. ARF latency ~2–4 weeks post-pharyngitis; APSGN ~1–2 weeks (throat) / 3–6 weeks (skin); RHD develops over years with recurrent infection.
- **Epidemiology (9):** GAS pharyngitis is among the most common outpatient infections; ~15–30% of pediatric sore throats. Not inherited (polygenic host susceptibility for sequelae). RHD >40.5M prevalent, ~306,000 deaths/yr (F012), concentrated in low/middle-income countries and Indigenous populations.
- **Diagnostics (10):** RADT/culture/molecular throat swab (F003); ASO & anti-DNase B serology (retrospective); C3 for APSGN (F011); Doppler echocardiography for ARF/RHD (F010). No genetic/omics diagnostics in routine use.
- **Prognosis (11):** Acute pharyngitis excellent (self-limited). RHD is the principal mortality driver. APSGN generally favorable short-term renal outcome in children.
- **Treatment (12):** Penicillin V / amoxicillin first-line (10-day course); benzathine penicillin G for adherence and secondary prophylaxis; macrolides/cephalosporins for penicillin allergy (F004). Supportive analgesia/antipyretics. No gene/cell/RNA therapies applicable.

- **Prevention (13):** Primary — treat pharyngitis to prevent ARF (F002); community programs (P 41956705).
Secondary — 4-weekly benzathine penicillin G prevents latent RHD progression (GOAL RCT, F012). No licensed vaccine; T-antigen/pilus and M-protein candidates in development (F009).
- **Other species / models (14–15):** GAS is a human-restricted pathogen; the **Lewis rat autoimmune valvulitis model** and mouse nasopharyngeal-challenge / rabbit immune-complex models are the principal experimental systems (F007, F009, F011).

Limitations and Knowledge Gaps

1. **No licensed human GAS vaccine.** T-antigen (TeeVax) and M-protein candidates show promise in animal models (F009), but human efficacy, safety (avoiding autoimmune cross-reactivity), and broad strain coverage remain unproven.
 2. **PANDAS/PANS remains contested.** The anti-D1R/anti-neuronal autoantibody mechanism and even the diagnostic entity are supported by translational and single-cell data (F008) but lack universally accepted, reproducible biomarkers.
 3. **Host-genetics effect sizes are modest.** HLA associations (OR $\approx$ 1.68 for DRB1*07) explain only part of susceptibility; GWAS in ARF/RHD are still emerging and underpowered outside a few populations (F005).
 4. **Quantitative ARF-prevention estimate rests on a single Cochrane analysis** with a wide CI (RR 0.22, 95% CI 0.02–2.08) because ARF is now rare in high-income trial settings (F008); most trials were not conducted where complication risk is highest.
 5. **Animal models are imperfect.** The Lewis rat reproduces valvulitis/carditis but not the full spectrum (chronic scarring, chorea, glomerulonephritis) of human disease, and no model recapitulates natural pharyngeal transmission-to-sequela progression.
 6. **Molecular-profiling gaps:** limited human transcriptomic/proteomic/metabolomic signatures of acute pharyngitis specifically (as opposed to invasive GAS or RHD tissue).
 7. **Strain–sequela specificity** (nephritogenic vs rheumatogenic emm types) is described but not mechanistically resolved.
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Proposed Follow-up Experiments / Actions

1. **Advance a multivalent GAS vaccine** combining T-antigen (TeeVax) and conserved M-protein epitopes toward controlled human infection / phase-appropriate trials, with rigorous screening for cardiac-myosin cross-reactivity before human dosing (extends F007, F009).
 2. **Single-cell + spatial transcriptomics of tonsillar tissue** during acute GAS pharyngitis vs viral pharyngitis, to map the epithelial-immune interface and superantigen-driven T/B-cell dynamics (extends F006).
 3. **Prospective HLA-genotyped cohort** linking pharyngitis episodes to ARF/APSGN outcomes to refine risk stratification (validate DRB1*07/DRB1*15 and DQ/IGHV4-61; F005) and enable precision secondary prophylaxis.
 4. **Define serum/CSF autoantibody panels (anti-cardiac-myosin, anti-D1R, anti-lysoganglioside)** with standardized assays to test PANDAS/Sydenham diagnostic reproducibility (F008).
 5. **Scale echocardiographic screening + benzathine penicillin G programs** in endemic regions, building on GOAL to define cost-effectiveness and adherence-support models (F012).
 6. **Molecular surveillance of M1UK and emm1/emm12 clones** with paired virulence-gene and antimicrobial-resistance profiling to anticipate scarlet-fever/invasive-disease surges (F006).
 7. **Nephritogenic-strain mechanistic studies** to identify the specific streptococcal antigen(s) (e.g., NAPlr, SpeB, IgG-binding proteins) driving immune-complex glomerulonephritis and to explain throat- vs skin-route differences (F011).
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Report compiled from 12 confirmed findings and 55 reviewed papers across 5 investigation iterations. Evidence quotes are verbatim from cited PubMed abstracts.
